

Non-Stationary Time Series Analysis of Temperature Data in India

Statistical Methods IV Project

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April 2026

Contents

- Introduction
- Statistical Methodology
- Data Description
- Exploratory Data Analysis
- Transforming towards Stationarity
- Model Selection
- Forecasting
- Result Summary
- Inference

Introduction

- Understanding long-term variations in temperature is central to studying climate dynamics and their broader societal implications.
- Temperature data typically exhibit strong **non-stationary behavior**, making them suitable for advanced time series modeling.
- In this project, we analyze the dataset “*Global Land Temperatures By Country*”, focusing on India.
- The dataset's long temporal span allows the study of both short-term seasonal variations and long-term climatic trends.
- Our main goal is to analyze the behaviour of temperature data in India using non stationary techniques and what insights can we draw from it.

Statistical Methodology

The analysis proceeds in a structured manner:

- ➊ **Data Cleaning:** Loading data and subsetting the data appropriately
- ➋ **Exploratory Data Analysis:** Summary statistics and plots to check for trend and seasonality.
- ➌ **Testing for Stationarity:** KPSS test to confirm non-stationarity.
- ➍ **Data Splitting and Transformation:** Train (80%) / Test (20%) split.
- ➎ **Model Identification and Fitting:** Using ACF and PACF plots to fit a SARIMA model and Ljung–Box test and RMSE for choosing our final model from our candidate models.
- ➏ **Forecasting and Validation:** Forecasting our predicted values using our fitted model and the compare the result with the real data.

Data Source

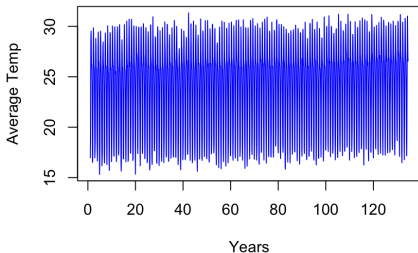
- Dataset: “*Global Land Temperatures By Country*”, publicly available on Kaggle.
- Contains monthly records of land temperatures for various countries starting from **1743**.
- Raw data: **577,462 observations** with 4 variables – Country, Date, Average Temperature, and Average Temperature Uncertainty.
- We restrict to **India** (1880–2013), reducing to **1604 observations** taking only **Average Temperature** as our key variable.
- The data is converted into a **time series object with frequency 12** to capture annual seasonal patterns.

Table: Summary Statistics of Temperature Data

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
15.32	20.13	25.46	24.09	27.23	31.33

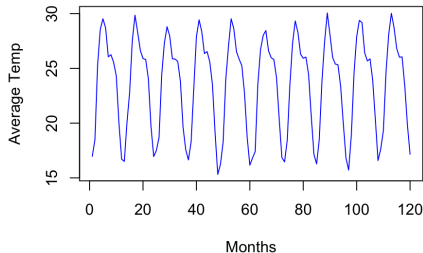
Time Series Plot

Average Temperature of India from 1880 to 2013



1880–2013

Average Temperature of India from 1880 to 1889



1880–1889

- We can see a **faint upward trend** suggesting a gradual increase in mean temperature.
- From the zoomed-in data we can see that there is a clear annual pattern suggesting that the data is non-stationary

KPSS Test for Stationarity

To formally assess stationarity, we employ the **KPSS (Kwiatkowski–Phillips–Schmidt–Shin)** test:

- H_0 : The time series is stationary (level stationary).
- H_1 : The time series is non-stationary (unit root / random walk component).

Test Statistic: Let $\hat{\eta}_t$ denote the residuals after regressing x_t on an intercept (and possibly a time trend). Then

$$\text{KPSS} = \frac{1}{n^2 \hat{\sigma}^2} \sum_{t=1}^n \left(\sum_{i=1}^t \hat{\eta}_i \right)^2,$$

where $\hat{\sigma}^2$ is an estimator of the long-run variance of $\{\eta_t\}$.

KPSS Test Result

Asymptotic Distribution: Under H_0 ,

$$\text{KPSS} \xrightarrow{d} \int_0^1 V^2(x) dx,$$

where $V(x)$ is a Brownian bridge.

Table: KPSS Test Result

Test	H_0	p-value	Conclusion
KPSS	Stationary	0.01	Non-Stationary

- Both graphical analysis and the KPSS test confirm that the data is **non-stationary**.
- This motivates the need for appropriate **transformations**.

Seasonal Mann–Kendall Test

The **Seasonal Mann–Kendall (SMK)** test detects a monotonic trend (H_1) in seasonal time series by applying the Mann–Kendall test within each season.

Let x_{ij} denote the observation in season $i = 1, \dots, m$ and year $j = 1, \dots, n_i$.

$$S_i = \sum_{k=1}^{n_i-1} \sum_{j=k+1}^{n_i} \text{sgn}(x_{ij} - x_{ik}), \quad S' = \sum_{i=1}^m S_i.$$

$$Z_{SMK} = \begin{cases} \frac{S' - 1}{\sqrt{\text{Var}(S')}}, & S' > 0 \\ 0, & S' = 0 \\ \frac{S' + 1}{\sqrt{\text{Var}(S')}}, & S' < 0 \end{cases}$$

SMK Test Result & Train–Test Split

Table: Result for Seasonal Mann–Kendall Test

Test	H_0	p-value	Conclusion
Seasonal Mann-Kendall	Constant Trend	1.15×10^{-72}	Trend exists

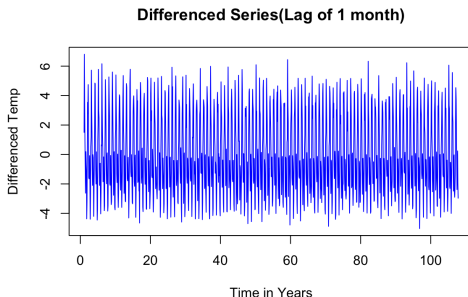
Train–Test Split:

- Chronological **80:20** split (preserves temporal order).
- Training set: **1283 months**.
- Testing set: **321 months**.

Step 1: Regular Differencing (Removing Trend)

We apply a first-order differencing to eliminate the trend component:

$$\nabla y_t = y_t - y_{t-1}$$



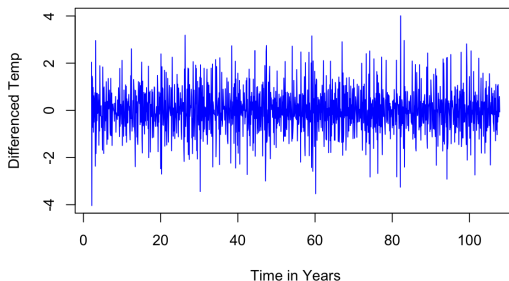
- Re-applying the SMK test gives a p-value of **0.49** $\Rightarrow$ fail to reject H_0 .
- The trend has been successfully removed.

Step 2: Seasonal Differencing (Removing Seasonality)

Apply seasonal differencing with lag 12 to the previously differenced series:

$$\nabla_{12}\nabla y_t = (y_t - y_{t-1}) - (y_{t-12} - y_{t-13})$$

Differenced Series(Lag of 12 months after a lag of 1 month)



- Both trend and seasonality are now removed.

Augmented Dickey–Fuller (ADF) Test

The ADF test checks for the presence of a **unit root**. It is based on the regression:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t$$

where the lagged difference terms account for higher-order autocorrelation.

Hypotheses:

- $H_0 : \gamma = 0$ (unit root, non-stationary)
- $H_1 : \gamma < 0$ (stationary)

The test statistic is $\tau = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})}$, which follows a non-standard Dickey–Fuller distribution under H_0 .

Confirmation of Stationarity

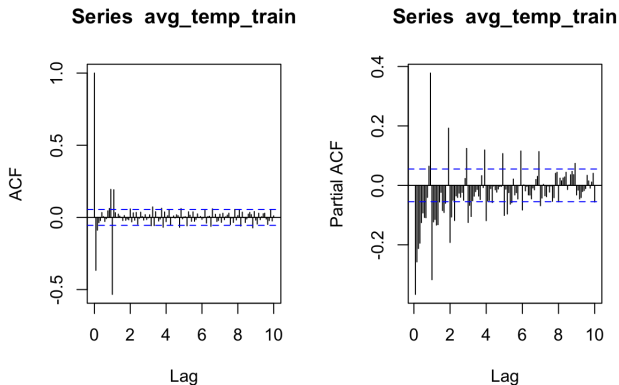
We apply both ADF and KPSS tests to formally confirm stationarity:

Table: Result of ADF and KPSS Test

Test	H_0	p-value	Conclusion
ADF	Unit Root	0.01	Stationary
KPSS	Stationary	0.10	Stationary

- The agreement between the two tests confirms the series has been successfully transformed into a **stationary process**.
- This justifies the use of **ARMA-type models** for subsequent analysis.

ACF and PACF of the Transformed Series



- A clear **seasonal MA(1)** component (spike at seasonal lag).
- ACF cuts off sharply after lag 1 or 2; PACF decays gradually.
- Non-seasonal component: **MA(1)** or **MA(2)** process.

ACF/PACF Pattern Guide

Pattern	Non-seasonal order	Seasonal order
ACF cuts off at lag q , PACF tails off	$MA(q)$	—
PACF cuts off at lag p , ACF tails off	$AR(p)$	—
Both tail off	$ARMA(p, q)$	—
Spike at lag 12 in ACF	—	$SMA(Q = 1)$
Spike at lag 12 in PACF	—	$SAR(P = 1)$

Candidate Models:

- $SARIMA(0, 1, 1)(0, 1, 1)_{12}$
- $SARIMA(0, 1, 2)(0, 1, 1)_{12}$

Benchmark and Model Selection

We fit both candidate models and compare using RMSE and Ljung–Box test.

Table: Box–Ljung Test Results

Model	χ^2	df	p-value
SARIMA(0, 1, 1)(0, 1, 1) ₁₂	62.13	12	9.205×10^{-9}
SARIMA(0, 1, 2)(0, 1, 1) ₁₂	13.43	12	0.3385

Table: RMSE of Residuals

SARIMA(0, 1, 1)(0, 1, 1) ₁₂	SARIMA(0, 1, 2)(0, 1, 1) ₁₂
0.6008	0.5864

- Model 2 has larger Ljung–Box p-value (residuals $\approx$ white noise) **and** lower RMSE.

Final Model Representation

We select $\text{SARIMA}(0, 1, 2)(0, 1, 1)_{12}$ as the final model:

$$(1 - B)(1 - B^{12})y_t = (1 + \theta_1 B + \theta_2 B^2)(1 + \Theta_1 B^{12})\epsilon_t$$

where B is the backshift operator, θ_1, θ_2 are the non-seasonal MA parameters, Θ_1 is the seasonal MA parameter, and $\{\epsilon_t\}$ is a white noise process.

Table: Parameter Estimates using `arima()`

ma1 (θ_1)	ma2 (θ_2)	sma1 (Θ_1)
-0.6631	-0.2317	-0.9700

Residual Analysis

- A well-fitted time series model should produce residuals that behave like **white noise**.
- Our chosen $\text{SARIMA}(0, 1, 2)(0, 1, 1)_{12}$ passes the Ljung–Box test ($p = 0.34 > 0.05$).
- Furthermore, the ACF plot of the residuals cuts off after lag 0, confirming the adequacy of the model.

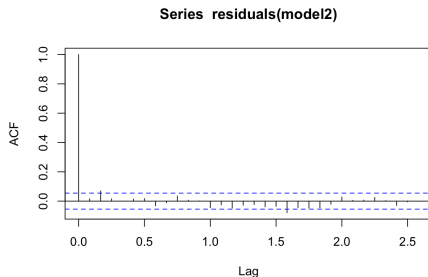


Figure: ACF of the residuals

Accuracy Metrics and Prediction Interval

- **Root Mean Square Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}$$

where e_t is the residual at time t .

- **Prediction Interval (95%):** Assuming normal errors,

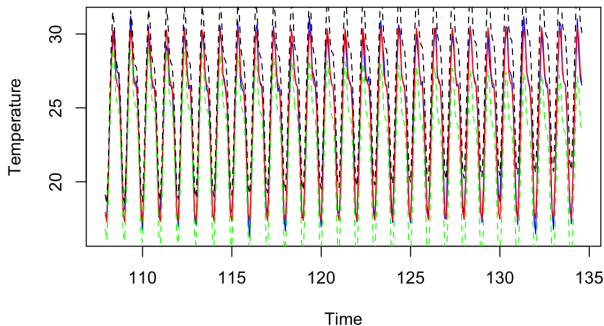
$$L_t = \hat{y}_t - 1.96 \hat{\sigma}, \quad U_t = \hat{y}_t + 1.96 \hat{\sigma}$$

$$\text{Coverage} = \frac{1}{n} \sum_{t=1}^n \mathbf{1}\{y_t \in [L_t, U_t]\}$$

- We obtain a coverage of **100%** $\implies$ the model captures future data well.

Forecast Plot

Predicted Avg Temperature of India from Dec 1986 to Aug 2013



- Predictions (red) align very closely with actual values (blue) from Dec 1986 to Aug 2013.
- Black and green dashed lines represent the 95% prediction interval bounds.

Results Summary

Stage	Decision / Finding
Data selection	India (1880–2013), 2613 observations
Time series construction	Monthly frequency 12
EDA	Clear seasonality and trend
KPSS (raw)	Non-stationary $\Rightarrow$ transform
Trend removal	First-order differencing ($d = 1$)
Seasonality removal	Seasonal differencing at lag 12 ($D = 1$)
Stationarity check	ADF + KPSS both confirm stationary
Final model	SARIMA(0, 1, 2)(0, 1, 1) ₁₂
Ljung–Box test	$p > 0.05 \Rightarrow$ white-noise residuals
Forecast horizon	321 months ahead
PI coverage	100% of actual values within bands

Inference

- The temperature series for India exhibits a **strong seasonal pattern** with periodicity of 12 months – corresponding months across years show similar temperature behavior.
- A **gradual upward trend** suggests a slow but persistent increase in average temperature, providing statistical evidence consistent with **global warming**.
- Through rigorous statistical techniques, we modeled this trend and seasonality using a **SARIMA**(0, 1, 2)(0, 1, 1)₁₂ model which provides accurate forecasts.

Thank you!